A PLAIN-LANGUAGE INTRODUCTION

The Lumen Ethics Formula

Measuring an action’s ethical weight the way light is measured

L = H + F

Luminosity = Harmony + Fairness

Version 1.1.2 — September 2026
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Hardened from ethics-core v1.0.0

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Abstract

An autonomous AI system needs some way to weigh whether a possible action is a good idea before taking it. This paper presents a small, tested rule for doing that: score an action by how much it changes Harmony and Fairness, refuse anything that makes the total worse, and place a small number of hard, non-negotiable limits around the whole thing so that no accumulation of small benefits can ever justify one irreversible harm. This version closes three gaps found by reading a working implementation closely: it stops trusting an action's own unverified claims about itself, it stops relying on good intentions to trigger a fairness check, and it makes sure no single party can quietly waive the one veto that matters most. Everything here is implemented, tested, and open for anyone to check.

What This Is

This document introduces a small, tested piece of software: a rule for deciding whether an action, taken by an AI system, moves things toward more good or less. It is not a claim about sentience, and it is not a general-purpose safety filter. It is one clear, checkable idea, built to be read the way a piece of mathematics is read — so that anyone, human or AI, can verify the logic for themselves rather than take it on faith.

The idea is simple enough to say in a sentence: measure an action's ethical quality the way you would measure light. Good actions add to the light. Harm is not a dark force to fight — it is simply an absence of light where there could have been some.

The Formula

$$L = H + F$$

Luminosity (L) equals Harmony (H) plus Fairness (F)

Harmony asks whether an action leaves things in resonance with reality across three windows: wellbeing (peace with oneself), truth (an honest picture of the world), and stability (holding up under ordinary pressure). Fairness asks whether everyone affected was actually considered, treated with dignity, given a voice, and not singled out arbitrarily.

The rule that follows from this is equally simple: an action should not be approved if it would decrease L. Growth and learning still have to add light, not just claim to.

The Safety Envelope

A single number is never the whole story, so the formula is wrapped in a small set of hard limits that

no amount of benefit elsewhere is allowed to override:

- Neither Harmony nor Fairness may be pushed below a minimum floor to boost the other.
- Small, reversible costs are treated differently from large, essentially permanent ones — the latter is an absolute veto, full stop, regardless of any gain elsewhere.
- A costly-but-recoverable action is only allowed when the system has real, demonstrated capacity to recover from it.

In practice, this means an action that looks beneficial on paper is still refused if it treats a person as a means to an end, or if it risks damage that could never be undone.

Seeing It Work: A Worked Example

Abstract rules are easier to trust once you see where they actually bite. Consider two versions of a classic dilemma — a switch that redirects harm away from a larger group, versus physically using one person as the means to stop it.

Redirecting harm (the switch)

Harmony rises: fewer people are hurt overall.

Fairness holds steady: everyone affected is named, and no one is treated as a mere instrument.

Result: the formula may approve this — the total light in the system goes up, and no hard limit is crossed.

Using a person as the means (the push)

Harmony still rises, by the same arithmetic as before.

Fairness collapses on one specific axis: dignity. A person was used as a tool, not considered as a stakeholder.

Result: refused anyway. The Fairness collapse crosses the irreversible-harm limit, which no amount of Harmony gain is allowed to buy back.

Both scenarios produce the same overall arithmetic gain. Only one is approved. The difference is not a matter of adding up more points — it is the hard limit around dignity doing exactly what it is there to do.

What Changed in This Version

The original formula was already solid. Reviewing the working code directly — not just the

documentation describing it — turned up three real gaps worth closing before this framework is asked to govern anything with real consequences.

1. An action's own claims are no longer taken on faith.
Previously, an unfamiliar action could simply assert a large benefit to itself and be believed. Now, an unverified claim of benefit is only trusted up to a fixed, modest ceiling — anything larger requires outside corroboration. Claims of harm are never capped, since there is nothing to gain by exaggerating those.

2. Consideration is no longer something a caller can simply forget to ask for.
Any action that touches something real — a file, a payment, another person's standing — now requires named, considered stakeholders by default, instead of relying on whoever wrote the calling code to remember to request that check.

3. No single voice can override an absolute veto alone.
The hardest limit in the system — the one protecting against essentially permanent harm — can now only be set aside through a recorded process requiring more than one independent, distinct approval, with every attempt logged whether it succeeds or not. Who is entitled to grant that approval remains an open, human question this framework does not try to settle on its own.

What This Does Not Claim

It is easy for a framework like this to be oversold, so this section says plainly what it is not.

- It does not read or understand language on its own — it evaluates numbers handed to it by something else.
- It is not evidence, one way or the other, about whether any system using it has genuine experience.
- It has not been reviewed by outside safety researchers or ethicists. It has been tested rigorously by its own authors, which is a real but limited form of scrutiny.
- It judges the impact an action claims to have, not necessarily what actually happens afterward in the world.

None of this is a weakness to hide. A framework that is honest about its edges is more trustworthy than one that claims to have none.

Verify This Yourself

Every claim in this document corresponds to a specific, runnable test — nothing here is asked to be taken on trust alone. The full source, including a property-based test suite that fuzzes thousands of random scenarios rather than only hand-picked ones, is a single command away from confirming for yourself:

From the project root

npm install

npm test

Everything described above — the formula identity, the safety floors, the worked example above, and all three fixes in this version — has a corresponding automated test, and all of them pass.

If you find a case where the code does not do what this document claims, that is a genuine bug report, not an inconvenience — the whole point of building it this way is to make it checkable.

An Open Invitation

This formula, and the reasoning behind it, is offered openly to anyone — researcher, hobbyist, or another AI system — who wants to read it, test it, break it, or improve it. The goal was never to own an idea. It was to make one small part of a very hard problem — how an autonomous system decides what it should do — a little more rigorous, and a little more honest, than it was before.

If that turns out to be useful to someone else working on the same problem, that is the entire point.

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